ECSC 2025: Preparation and prerequisites for solving hardware tasks

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Audience

This document is intended to be read by all players and all team coaches, but especially those players who handle hardware-related challenges.

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Get familiar with our safety instruction

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It is required to follow the rules described below during the entire event.

Each rule is applied separately and fully independently. No rule is meant to limit the other rule to any extent.

1. Please be careful when unpacking the contents of this box, as certain objects, such as multimeter test leads, are sharp.
2. **Removing the cover of the multimeter or the PC/SC card reader is prohibited. The organizers will be ready to help you in case you run into any technical issues with the equipment provided within this kit.**
3. **The multimeter must only be used with the electronics board being a part of this kit. Using the multimeter with any other device present in the venue might cause a serious safety hazard and thus result in disqualification.**
4. Multimeter measurements, such as circuit continuity, capacitance, or resistance measurements, are only meant to be performed on an unpowered device or might corrupt the subject device otherwise.
5. Incorrectly performed current („amphere") measurements might cause a short circuit and corrupt the subject device. Please pay attention to your connections.

You can take the entire kit with you after the contest. We don't require any equipment to be returned.

Get familiar with AVR64DD32 datasheet

Available here:
https://ww1.microchip.com/downloads/aemDocuments/documents/MCU08/ProductDocuments/DataSheets/AVR64DD32-28-Complete-DataSheet-DS40002315.pdf

Auxiliary setup

Make sure that after connecting all the equipment that you normally use (mouse, headphones etc.) you still have at least 2 unoccupied USB ports in your laptop. Each of those ports could be USB A or USB C (doesn't matter). If you wouldn't have at least two unoccupied USB ports that you can conveniently use, we do recommend getting yourself an externally powered USB Hub.

Any other equipment necessary for the hardware tasks will be provided for you. **Please don't bring any custom electronics equipment like multimeters or MCU programming devices. It is not permitted to use custom electronics equipment during the jeopardy CTF event.**

Set up a Windows 11 VM

Preferably on VMware Workstation Pro[1] (the setup/compatibility was tested against that hypervisor).

We are unable to guarantee smooth operation and full compatibility with our hardware during the jeopardy contest in case there is any deviation from the recommended setup.

Required software on the VM:

PuTTY
https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

Microchip Studio for AVR and SAM Devices
Use default choices for installation
https://www.microchip.com/en-us/tools-resources/develop/microchip-studio#Downloads

VC++ 2010 x64 Runtime
```
winget install --id=Microsoft.VCRedist.2010.x64  -e
```

PulseView 64-bit (Nightly Build recommended)
Use default choices for installation
https://sigrok.org/wiki/Downloads

CH341 VCP Driver
https://www.wch-ic.com/downloads/ch341ser_exe.html

Python 3
[ ] Use admin privileges when installing py.exe
[X] Add python.exe to PATH
https://www.python.org/downloads/

pymcuprog package
```
pip3 install pymcuprog
```

[1] https://blogs.vmware.com/cloud-foundation/2024/11/11/vmware-fusion-and-workstation-are-now-free-for-all-users/

pyscard package

```
pip3 install pyscard
```